



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



УНИВЕРЗИТЕТ У БЕОГРАДУ
UNIVERSITY OF BELGRADE

X-Ray Emitting Plasma in Galaxies and Galaxy Clusters

SPECTROSCOPY OF ASTROPHYSICAL PLASMAS

Fritz Ali Agildere

June 9, 2026



Left to Right, Top to Bottom: Abell 85, Abell 426 (Perseus), Abell 1644, Abell 1656 (Coma), Abell 2029, and Abell 3158.



Left to Right, Top to Bottom: Abell 85, Abell 426 (Perseus), Abell 1644, Abell 1656 (Coma), Abell 2029, and Abell 3158.

Overview

Physical Properties

- Galaxy Clusters
- Radiation Mechanisms

Radial Profiles

- Data Processing
- Theory & Results

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- Detectors & Telescopes
- Spectral Characteristics
- Conclusion & Outlook

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Cluster Models & Classes

Cluster Models & Classes

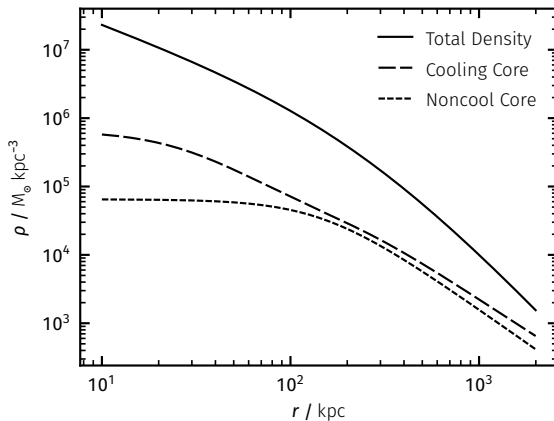
- Purely Gravitational Baseline Assumption

Cluster Models & Classes

- Purely Gravitational Baseline Assumption
- Dark Matter Halo Dominated Systems

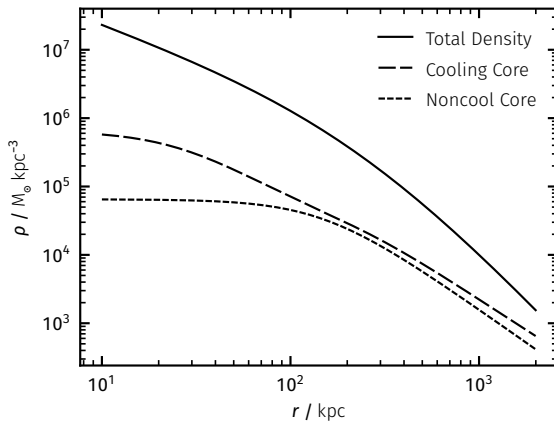
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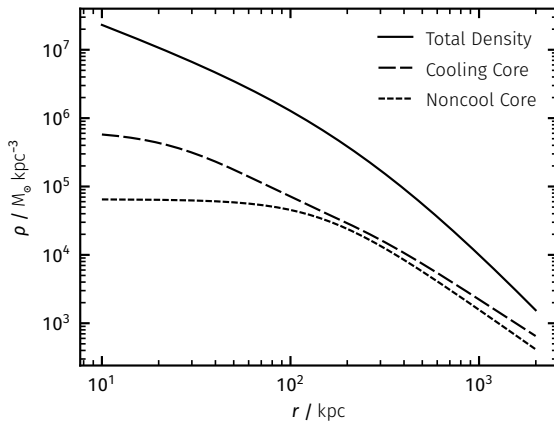
Cluster Models & Classes

- Purely Gravitational Baseline Assumption
- Dark Matter Halo Dominated Systems
- Selfsimilar Radial Scaling



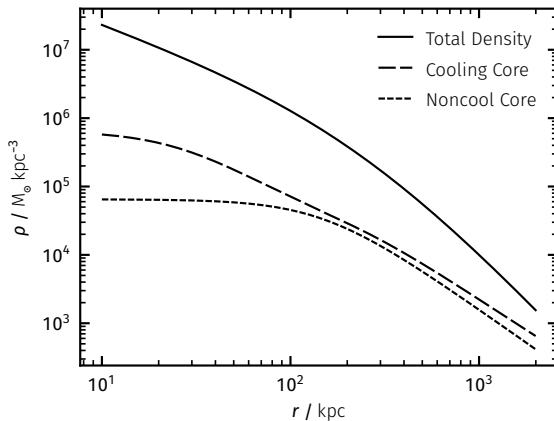
Cluster Models & Classes

- Purely Gravitational Baseline Assumption
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- Selfsimilar Radial Scaling
- Deviation Defines Categories



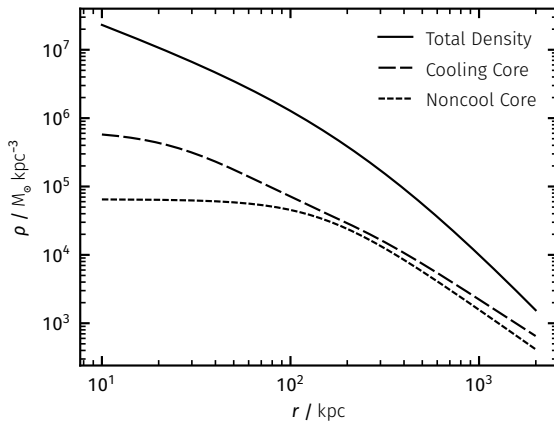
Cluster Models & Classes

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 - Cooling Cores (CC)



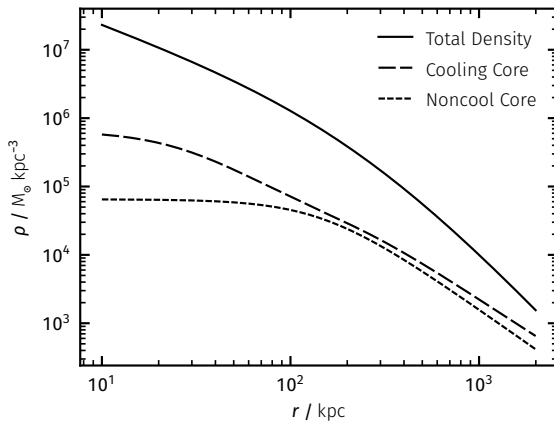
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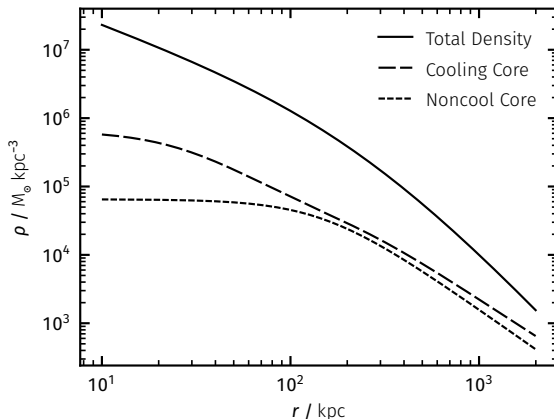
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- Mergers Inject Energetic Shocks & Sloshing
- Central Active Galactic Nuclei (AGN) Feedback
Effects Prevent Total Collapse



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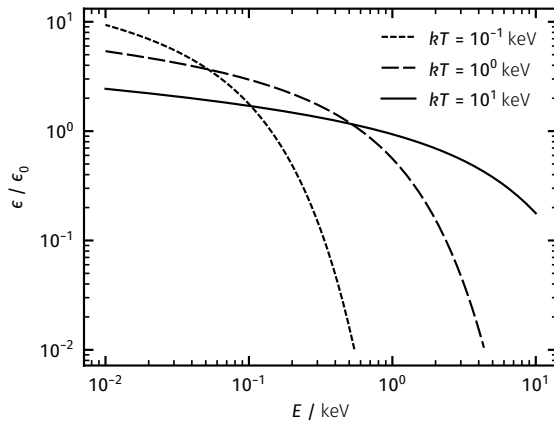
Characteristic Emission

Characteristic Emission

- Bremsstrahlung (free-free)

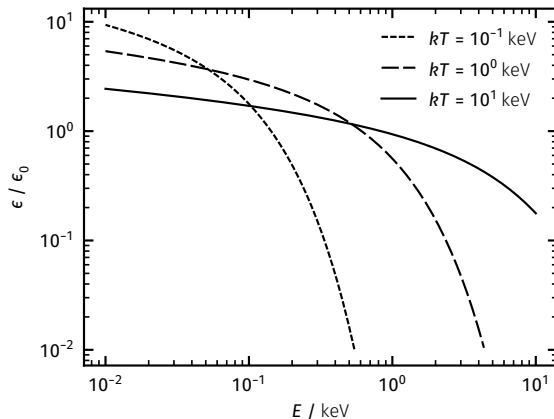
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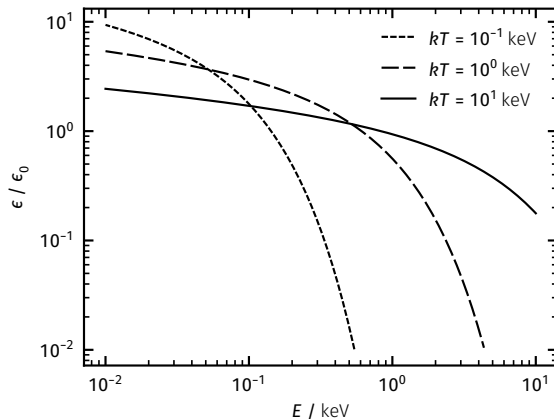
Characteristic Emission

- Bremsstrahlung (free-free)
- Broad Continuous Spectrum



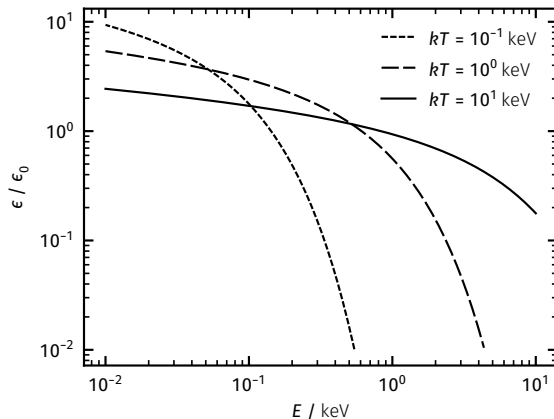
Characteristic Emission

- Bremsstrahlung (free-free)
 - Broad Continuous Spectrum
- Recombination (bound-free)



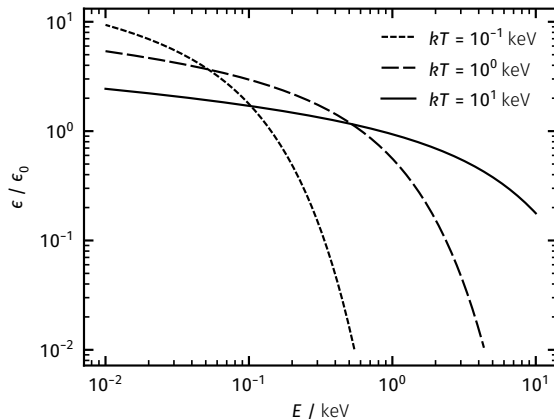
Characteristic Emission

- Bremsstrahlung (free-free)
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- Recombination (bound-free)
 - Soft Continuum



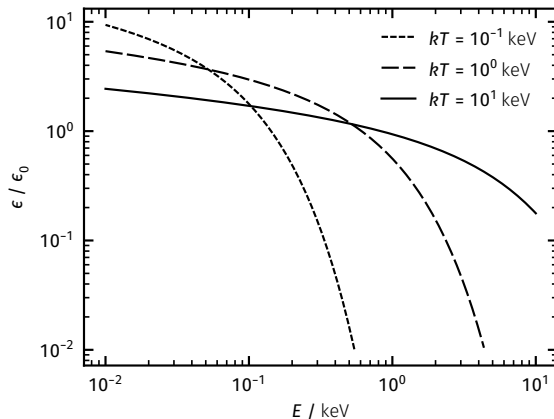
Characteristic Emission

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 - Soft Continuum
 - Hard Ionization Edge



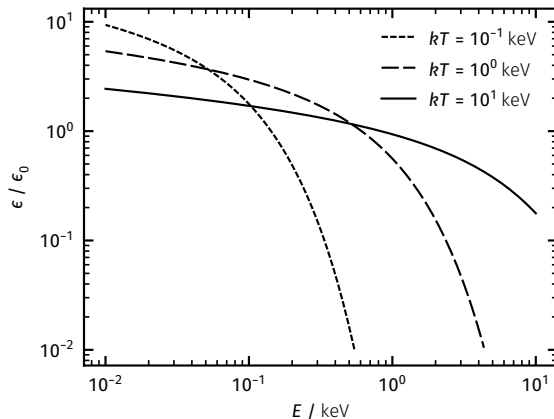
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- Atomic Transitions (bound-bound)



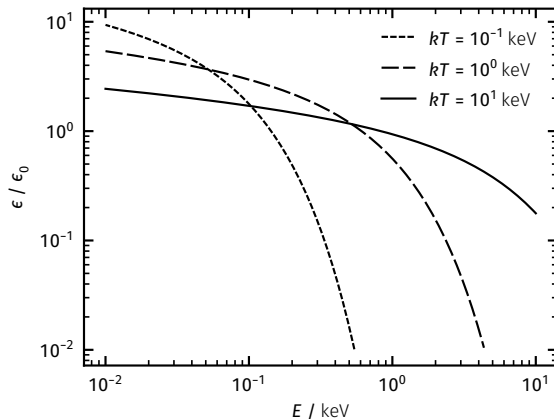
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 - Discrete Lines



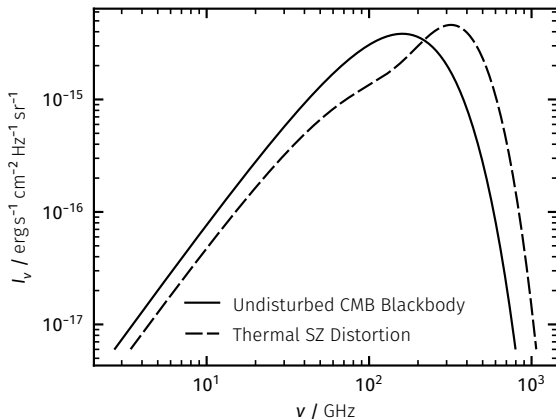
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- Sunyaev-Zeldovich Effect (Inverse Compton)



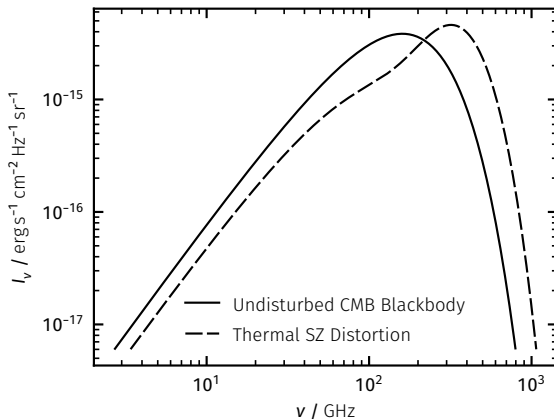
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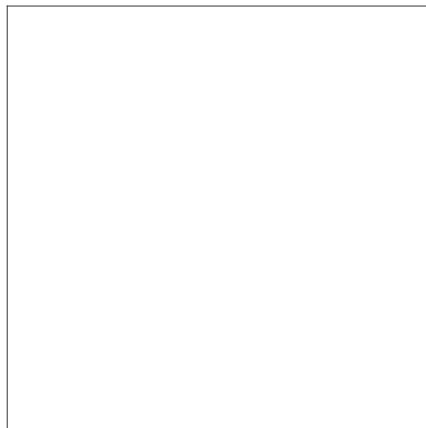
Extracting Information

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- Annulus Decomposition

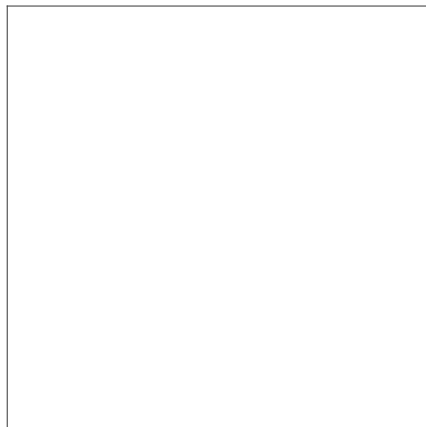
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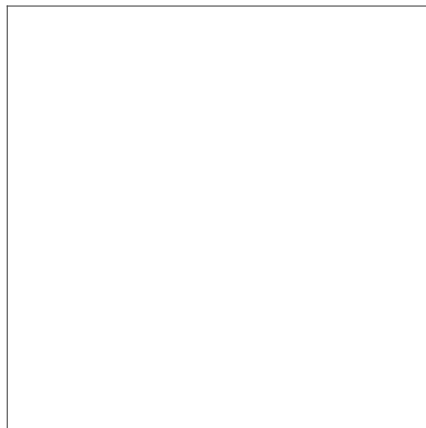
Extracting Information

- Annulus Decomposition
- Projection Unfolding



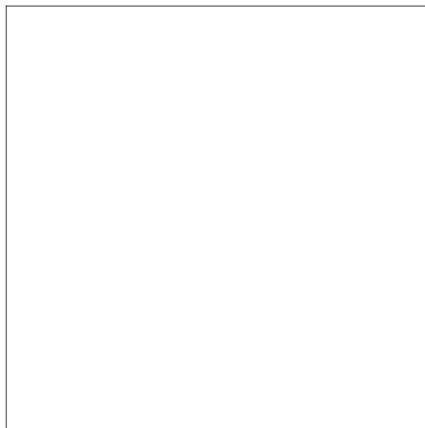
Extracting Information

- Annulus Decomposition
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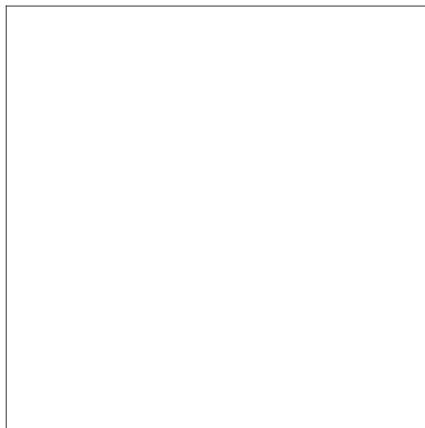
Extracting Information

- Annulus Decomposition
- Projection Unfolding
- Spectral Fitting
 - Obtain Temperature, Density, Abundances, Redshift



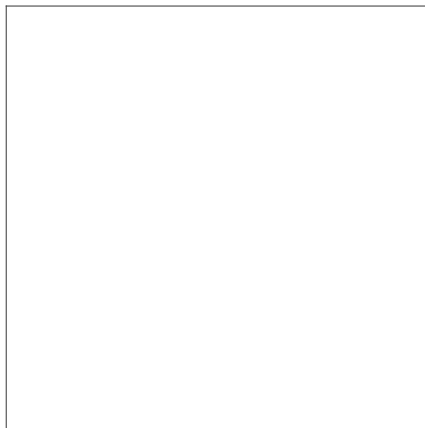
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 - Derive Pressure, Entropy, Mass



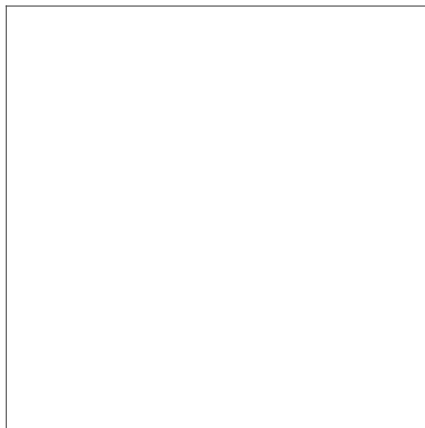
Extracting Information

- Annulus Decomposition
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 - Obtain Temperature, Density, Abundances, Redshift
 - Derive Pressure, Entropy, Mass
- Spherically Symmetric Averaging



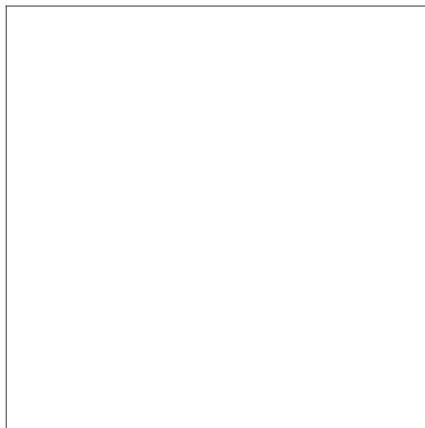
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- Spherically Symmetric Averaging
- Disturbed Systems Breakdown Possible



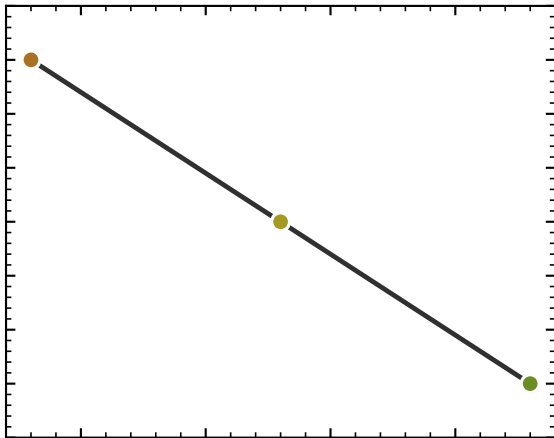
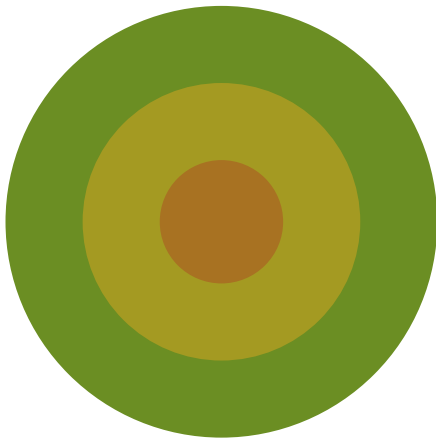
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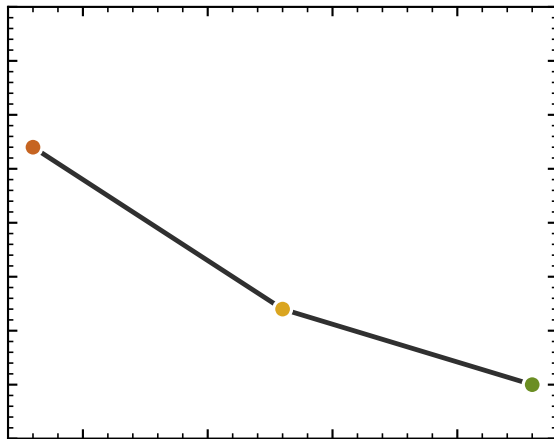


Deprojection

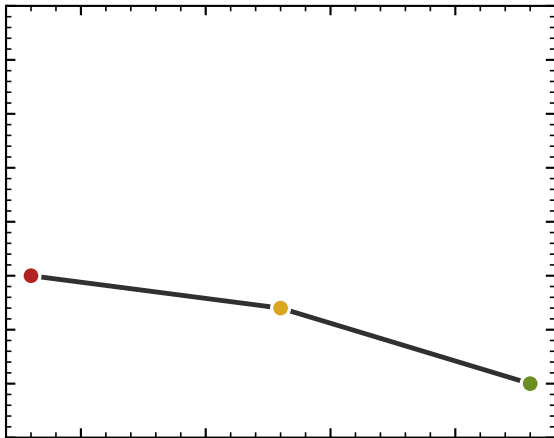
Deprojection



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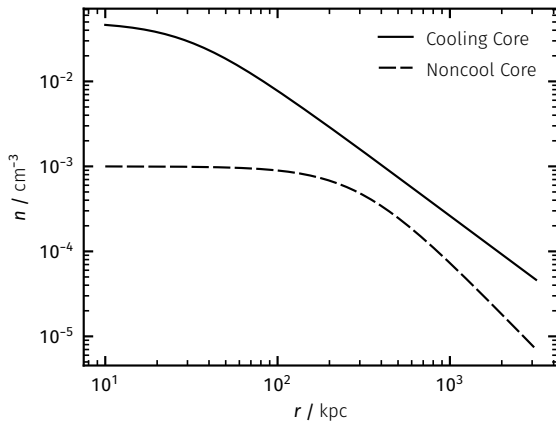
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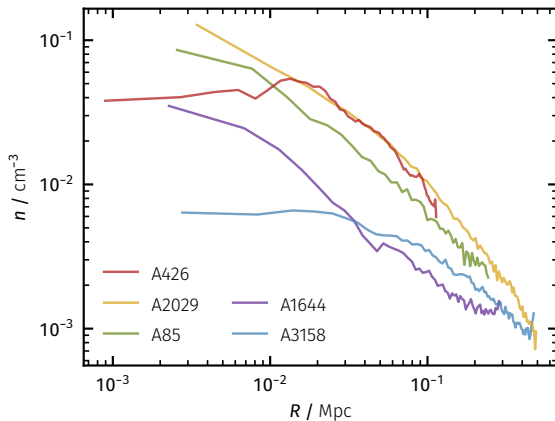
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Density & Pressure

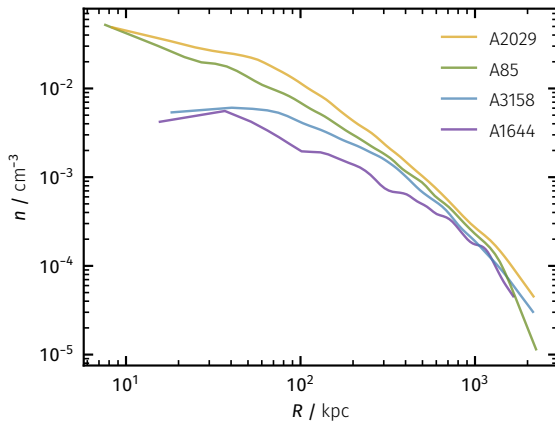
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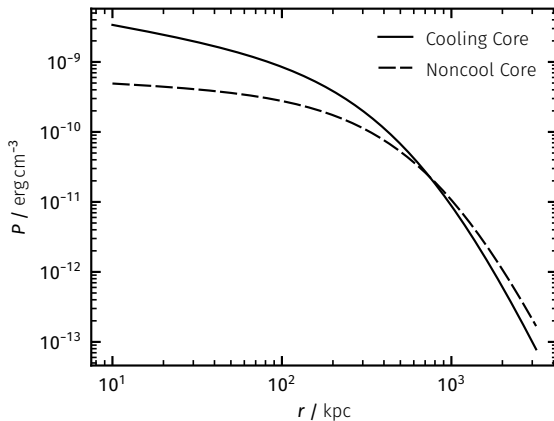
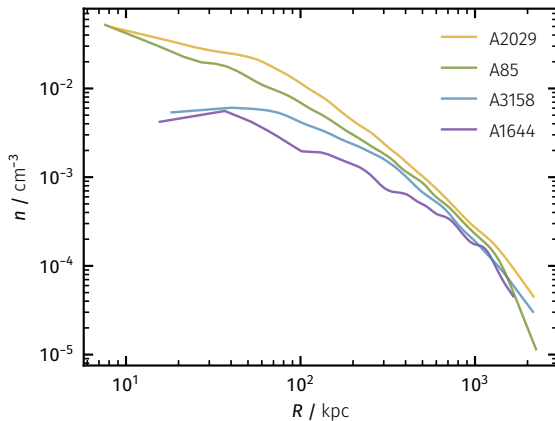
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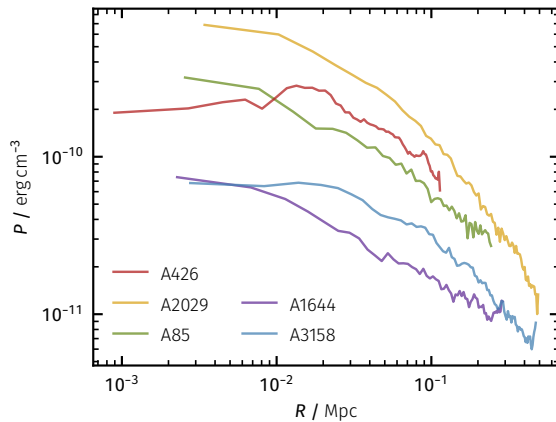
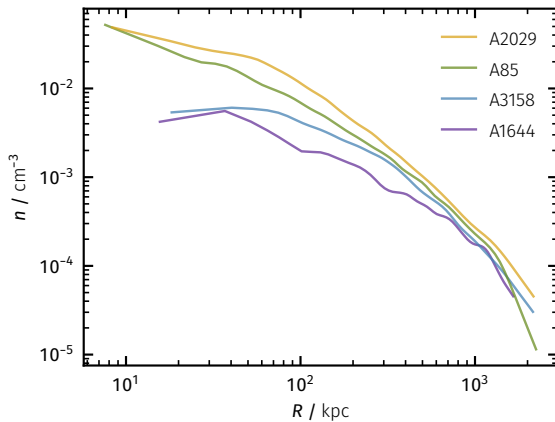
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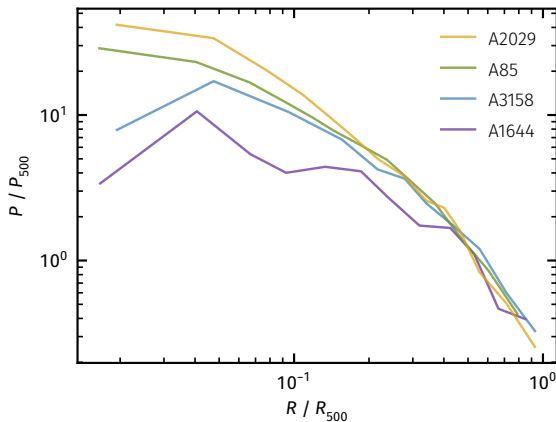
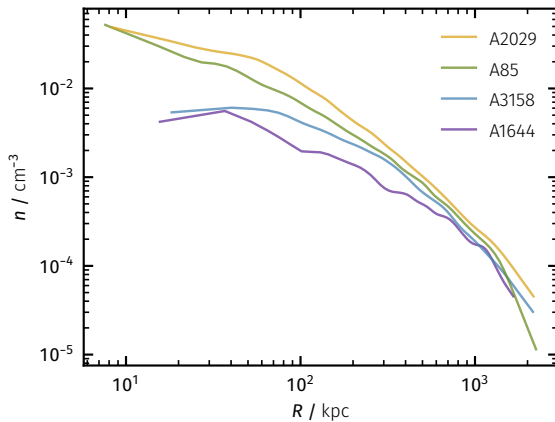
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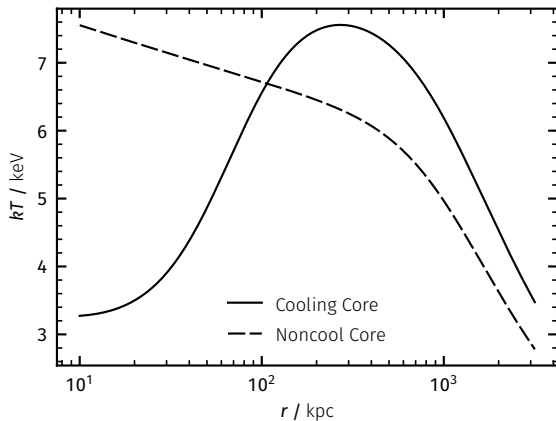


Density & Pressure

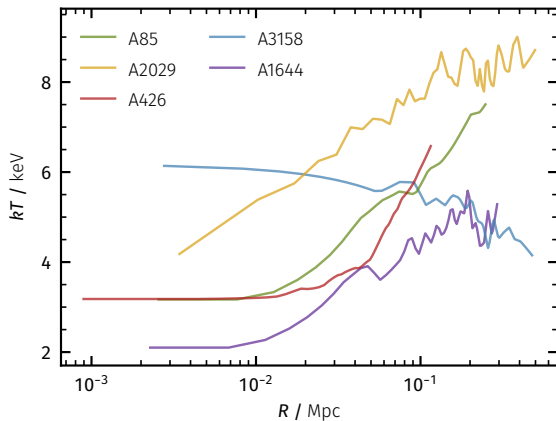


Temperature & Entropy

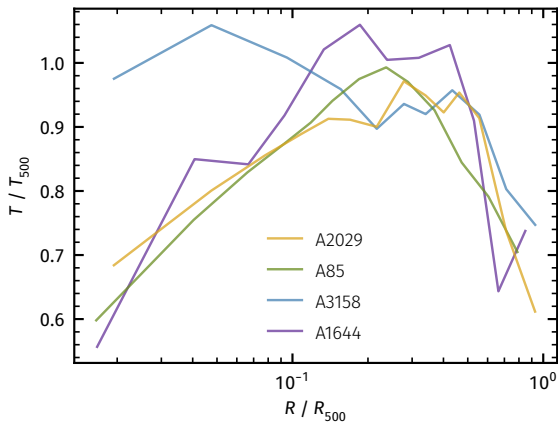
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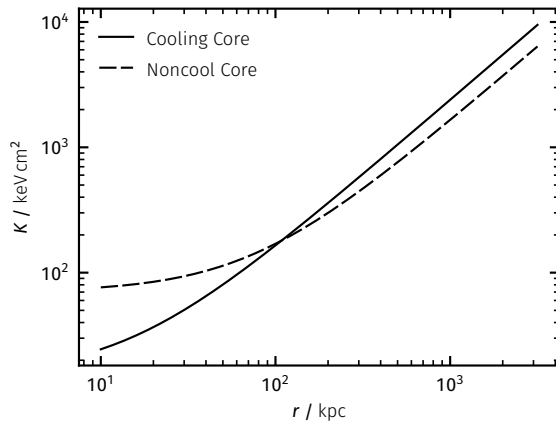
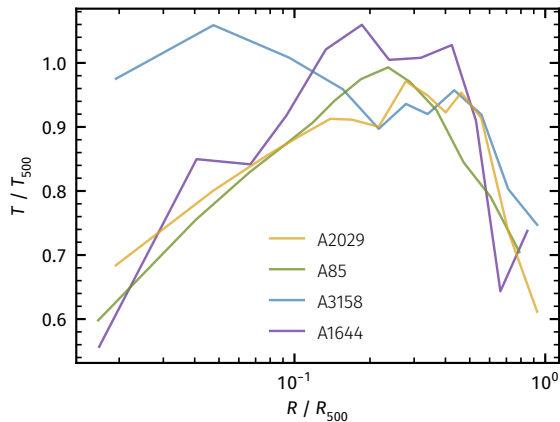
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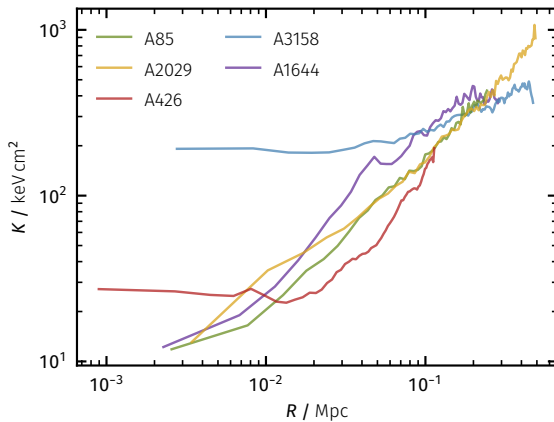
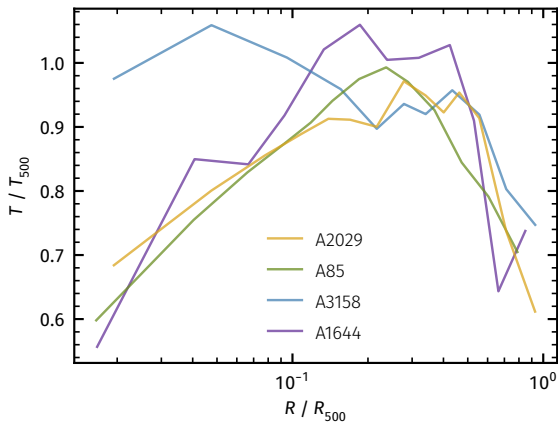
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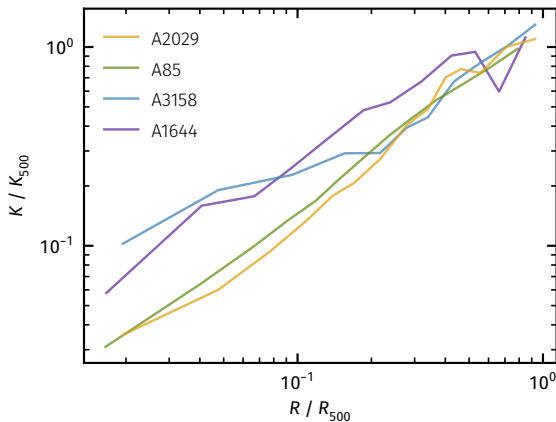
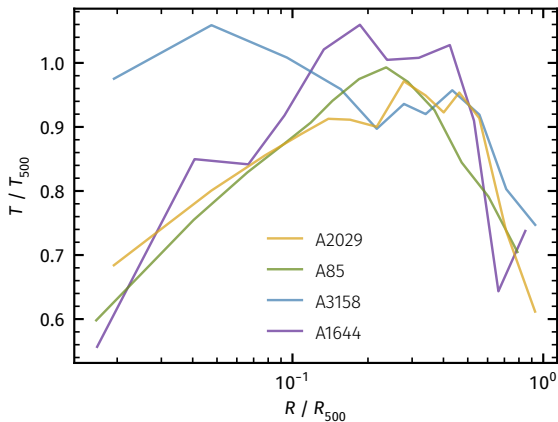
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- Large Survey

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 - ROSAT (retired)

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- Large Survey
 - ROSAT (retired)
 - Planck (retired)

Selected Missions

- Large Survey
 - ROSAT (retired)
 - Planck (retired)
 - eROSITA (active)



Selected Missions

- Large Survey
 - ROSAT (retired)
 - Planck (retired)
 - eROSITA (active)
- Spectroscopic Precision



Selected Missions

- Large Survey
 - ROSAT (retired)
 - Planck (retired)
 - eROSITA (active)
- Spectroscopic Precision
 - ASCA (retired)



Selected Missions

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 - XMM-Newton (active)



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 - XRISM (active)



Selected Missions

- Large Survey
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 - Planck (retired)
 - eROSITA (active)
- Spectroscopic Precision
 - ASCA (retired)
 - XMM-Newton (active)
 - XRISM (active)
- Spatial Resolution



Selected Missions

- Large Survey
 - ROSAT (retired)
 - Planck (retired)
 - eROSITA (active)
- Spectroscopic Precision
 - ASCA (retired)
 - XMM-Newton (active)
 - XRISM (active)
- Spatial Resolution
 - Chandra (active)



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 - XRISM (active)
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 - Chandra (active)
 - NewAthena (planned)



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Discussion

Thank You!

- R. C. Hickox & D. M. Alexander, *Obscured Active Galactic Nuclei* (2018) ARA&A Volume 56
- P. F. Hopkins et. al., *Stellar and Quasar Feedback in Concert: Effects on AGN Accretion, Obscuration and Outflows* (2016) MNRAS Volume 458 Issue 1

Resources on GitHub: [here](#)

Appendix